

# 1.3 Components of HCI

ایچ سی آئی کے چار بنیادی اجزاء ہیں: صارف (اینڈ یوزر جو خود سسٹم چلاتا ہے اور اسٹیک ہولڈر جس کا نتیجے میں مفاد...

CS XII

Chapter 1: Computer Systems (HCI)

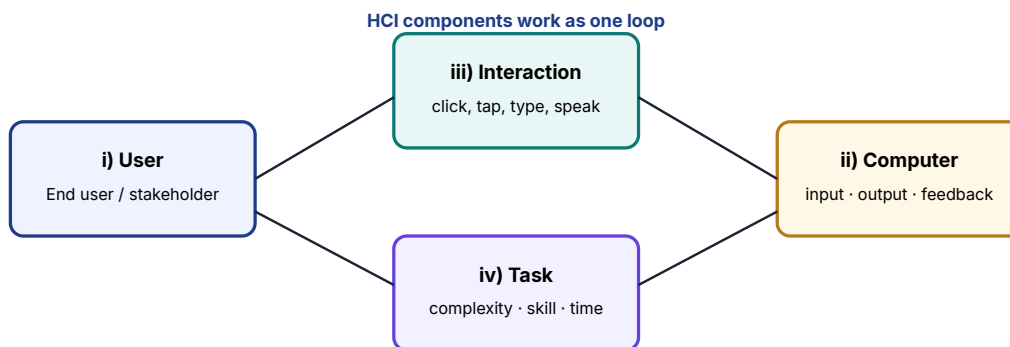
Lecture 04 of 15

## 1.3 Components of HCI

### LEARN IT

HCI relies on several primary components that work together. If one component is weak, the whole interaction feels broken. The first four are the people, the machine, the conversation, and the job to be done.

|     | Component                               | Meaning                                                                                                                                                                                                                                                                                                      |
|-----|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| i   | <b>User</b> (صارف)                      | Any individual or group that interacts with a computer system to achieve defined goals. Two categories: <b>end users</b> (people who directly operate the system, from beginners to experts) and <b>stakeholders</b> (people with an interest in the outcome, such as managers and clients).                 |
| ii  | <b>Computer system</b> (کمپیوٹر سسٹم)   | The platform where humans and computers communicate through input, output and feedback using an interface. It connects human actions with machine responses.                                                                                                                                                 |
| iii | <b>User interaction</b> (صارف کا تعامل) | The process through which a user communicates with a system by performing actions (clicking, tapping, typing, speaking). It transforms user intent into system responses.                                                                                                                                    |
| iv  | <b>Task</b> (کام)                       | The specific action users perform (for example, booking a ticket). It has three characteristics: <b>task complexity</b> (filling a form vs designing a 3D model), <b>user knowledge</b> (basic typing vs advanced programming), and <b>time requirement</b> (ATM withdrawal vs online passport application). |



The user performs a task through interaction with a computer system. All four parts must fit together.

### WORKED EXAMPLE

#### Worked example: label the four components of an HBL ATM withdrawal.

1. **User:** end user = the account holder standing at the machine; stakeholder = the bank that owns the ATM and must keep it safe and working.
2. **Computer system:** the ATM hardware plus the bank's software; it takes PIN input, shows a screen, and dispenses notes (output and feedback).
3. **User interaction:** insert card → type PIN → choose "Cash" → type amount → confirm. Intent ("I want Rs 5,000") becomes a machine response (cash + receipt).
4. **Task:** withdraw cash. Complexity: low (a few buttons). User knowledge: basic (PIN and amounts). Time: about one minute — much shorter than an online passport application.

### REMEMBER IT

**U-C-I-T:** User, Computer system, Interaction, Task. "U Can Interact with Tasks." Then add three task questions: How hard? What skill? How long?

### COMMON MISTAKE

**Mistake:** Calling the bank manager an "end user" of the ATM, or saying a task has only one property (difficulty).

**Fix:** The manager is usually a **stakeholder**. The person pressing the keys is the **end user**. A task always has complexity, knowledge and time — mention all three in long answers.

### اردو میں سمجھیں

ایچ سی آئی کے چار بنیادی اجزاء ہیں: صارف (اینڈ یوزر جو خود سسٹم چلاتا ہے اور اسٹیک ہولڈر جس کا نتیجے میں مفاد ہے)، کمپیوٹر سسٹم (ان پٹ، آؤٹ پٹ اور فیڈبیک کا پلیٹ فارم)، صارف کا تعامل (کلک، ٹیپ، ٹائپ، بولنا) اور کام (ٹاسک) جس کی پیچیدگی، مطلوبہ مہارت اور وقت مختلف ہوتا ہے۔ اے ٹی ایم سے رقم نکالنا آسان اور مختصر کام ہے؛ پاسپورٹ کی آن لائن درخواست لمبا کام ہے۔

### CHECK YOURSELF

1. What is the difference between an end user and a stakeholder?
2. Name the three characteristics of a task.
3. In a school portal, who is a stakeholder?

Answers: 1. End users directly operate the system; stakeholders have an interest in the outcome (managers, clients) but may not click the buttons. 2. Task complexity, user knowledge, time requirement. 3. The principal, IT admin or board office — they care about results even if students are the end users.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** The user in HCI is:

- a. The human with goals, limits, skills
- b. The CPU
- c. The database
- d. The sprint log

**MCQ 2.** The task is:

- a. What the user is trying to get done
- b. The Wi-Fi password
- c. The NAND count
- d. The pie flavour

**MCQ 3.** The context/environment is:

- a. Where/when/noise/light/stress
- b. Only RGB of the logo
- c. Only Big O
- d. Only SQL

**MCQ 4.** Missing any component in design causes:

- a. HCI problems
- b. Faster CPUs
- c. Better keys
- d. Shorter OSI

**MCQ 5.** A tired nurse at night is part of:

- a. User + environment
- b. Only the algorithm
- c. Only the DBMS
- d. Only Figma auto-layout

**MCQ 6.** Components work:

- a. Together
- b. In isolation always
- c. Only in waterfall phase 6
- d. Only after MVPs ship

## Extra short questions

1. List the primary HCI components from the notes.
2. Give a college-lab example of each.
3. What if you design only for an expert user?
4. Why environment?

## Extra long question

1. Analyse JazzCash (or a similar app) using HCI components. Where might it fail for a first-time Urdu user in bright sunlight?

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

### Short

1. User, task, interface, environment/context, feedback (and system).
2. User: XI student. Task: submit practical. UI: LMS upload. Env: noisy lab, short period. Feedback: "uploaded" tick.
3. Beginners fail — a typical HCI problem.
4. A UI that needs silence fails in a crowded canteen kiosk.

### Long

1. User literacy/language; task send-money; UI contrast; outdoor glare; feedback of "success". Recommend larger type, Urdu toggle, confirmation screen.